Multiple Choice Question - 5

B **1. What is the main purpose of analog-to-digital conversion?**

- A) To amplify analog signals
- B) To convert analog data into digital signals
- C) To increase data bandwidth
- D) To decode digital signals

C **2. Which device is used to convert analog data to digital and vice versa?**

- A) Modem
- B) Multiplexer
- C) Codec
- D) Repeater

B **3. What is the Nyquist rate for a signal with a maximum frequency of 4000 Hz?**

- A) 4000 samples per second
- B) 8000 samples per second
- C) 2000 samples per second
- D) 1000 samples per second

sampling rate ≥ 2×fmax

A **4. Which quantization method assigns the middle value of each zone as the quantized value?**

- A) Pulse code modulation
- B) Delta modulation
- C) Linear quantization
- D) Non-linear quantization

A **5. In polar Non-Return-to-Zero Level (NRZ-L) encoding, what represents a binary 0?**

- A) Positive voltage
- B) Zero voltage
- C) Negative voltage
- D) Alternating voltage

A **6. What is the main disadvantage of Return-to-Zero (RZ) encoding?**

- A) Complexity and higher bandwidth requirements 因为RZ有高
- B) Low error tolerance
- C) Absence of DC component
- D) Lack of self-synchronization

C

7. Which encoding technique is specified for the IEEE 802.3 Ethernet standard?

- A) Non-Return-to-Zero (NRZ)
- B) Return-to-Zero (RZ)
- C) Manchester encoding
- D) Bipolar-AMI

B

8. In digital transmission, what is the signal rate for NRZ-I with a data rate of 10 Mbps?

- A) 20 kbaud
- B) 5 Mbaud
- C) 10 Mbaud
- D) 2.5 Mbaud

D

9. Which transmission mode is most suitable for multimedia applications, such as audio and video?

- A) Parallel
- B) Asynchronous
- C) Synchronous
- D) Isochronous

C

10. Which of the following describes block coding?

- A) Converts each bit to a symbol
- B) Converts analog to digital signals
- C) Divides bits into blocks for improved error detection
- D) Eliminates quantization error

C

11. In 4B/5B block coding, how many bits are used to replace every 4-bit input?

用5bit替換4bit

- A) 3
- B) 4
- C) 5
- D) 6

C/D

12. What problem does 4B/5B block coding solve in NRZ-I encoding?

- A) High bandwidth requirements
- B) Presence of a DC component
- C) Long sequence of 0s causing synchronization issues
- D) Lack of self-synchronization

C

13. In delta modulation, what does a '1' bit indicate?

- A) No change in amplitude

- B) Decrease in amplitude
- C) Increase in amplitude
- D) Error correction

C **14. Which digital-to-digital encoding technique has no DC component and self-synchronizing capability?**

- A) NRZ-I
- B) Unipolar NRZ
- C) Differential Manchester
- D) AMI